

CLASS 4 DECOMMISSIONING COST ESTIMATES
For
The Australian Nuclear Science and Technology Organisation
(ANSTO)

(Buildings B17, B70, B18, B54, & B56)

March 21, 2026



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REVISION HISTORY

Revision	Issue Date	Summary of Changes	Changes by
0	March 21, 2026	Initial Issue after ANSTO comments incorporated	JEC

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Acronym List

ANSTO	Australian Nuclear Science and Technology Organisation
ACM	Asbestos Containing Material
CAA	Carignan & Associates, LLC
DCE	Decommissioning Cost Estimate
DGC	Decommissioning General Contractor
DOE	Department of Energy
PT&C	Project Time and Cost
ROM	Rough Order of Magnitude
UCF	Unit Cost Factor
US	United States
WBS	Work Breakdown Structure
WDF	Work Difficulty Factor

EXECUTIVE SUMMARY

In 2018, the Australian Nuclear Science and Technology Organisation (ANSTO) Contracted with Project Time and Cost to prepare a site-wide decommissioning and dismantlement (D&D) Class 4 estimate for all areas and structures on the Lucas Heights Site. The 2018 estimate was prepared using PT&C proprietary software and approach. When it was time to update the estimate in 2025, PT&C no longer provided this service, software, or the support to update the estimates they had previously prepared (Ref 6). To address this issue, ANSTO contracted with Carignan and Associates LLC (CAA) to provide estimates on a select number of structures on the ANSTO site. In addition, CAA was tasked with providing a cost estimating template that would allow ANSTO to update their estimates in house with little or no contractor support.

To accomplish this, ANSTO selected five facilities for CAA to prepare decommissioning cost estimates: B17, B70, B56, B54 and B19. The building/areas were selected to represent the types of facilities on the ANSTO site. CAA prepared level IV estimates, often referred to as rough-order-of-magnitude (ROM) estimates for each structure.

The estimates were prepared using information from existing United States decommissioning cost estimates CAA estimators had already developed. This information, along with information from ANSTO staff, the PT&C 2018 estimate, and estimator experience was used in developing these estimates. The estimates prepared served two purposes; Develop the level four estimate for each structure and provide a method and approach that would allow ANSTO to update their estimates in house with little or no contractor support.

The five decommissioning cost estimates follow industry recognized cost estimating methods and best practices from the American Association of Cost Engineers International (AACEI), and guidance provided in the Department of Energy (DOE) guide DOE G 413.3-21A, Cost Estimating Guide. Accordingly, a graded approach was used based on limited information available during preparation of the estimate. Due to limited site information such as a lack of characterization data, a Historical Site Assessments, and minimal building information such as inventory, these estimates are considered Class IV estimate with a range of accuracy between -30% and +50% as shown below in Table 1.

The DCE for each facility is shown in table 1.

Facility	Description	Cost 2026 AUD
B17	Material Storage & Workshop	\$3,021,528
B18	Remediation Work - Mineral Sands	\$5,178,910
B54	Mo99 Processing Facility	\$14,159,309
B56	Institute of Materials & Engineering Science	\$14,150,956
B70	HIFAR Electrical Power Supply	\$765,187
Total		\$37,275,890

Table 1 - Summary of Costs

1. INTRODUCTION/BACKGROUND

In 2018, ANSTO Contracted with Project Time and Cost to prepare a site-wide decommissioning and dismantlement (D&D) Class IV estimate for all areas and structures on the Lucas Heights Site. The estimate was prepared using PT&C company proprietary software and approach. When it was time to update the estimate in 2025, PT&C no longer provided this service or support. Without PT&C proprietary software and the knowledge to operate it, ANSTO was unable to prepare a decommissioning cost update.

To address this issue, ANSTO contracted with Carignan and Associates LLC (CAA) to develop a Decommissioning Cost Estimate (DCE) format and approach that would allow ANSTO to prepare class IV DCE updates in house with little or no contractor support. To accomplish this, ANSTO identified five facilities for CAA to prepare DCEs. The five buildings ANSTO selected were B17, B70, B18, B54, & B56. All are located on the Lucas Heights Site. Using this site data, CAA prepared a DCE for each building. The estimates were prepared in a manner which allows ANSTO to prepare updates in-house with minimal contractor support.

The format or template developed allows ANSTO to change key parameters, including site-specific data such as labor rates, crew size, escalation rates, or specific waste costs. Changing these key parameters for a specific facility will automatically adjust the Class 4 estimate.

Due to the limited amount of site-specific data and without a site visit, the estimates provided in this report are considered class IV estimates. A class IV estimate as defined by the Association for the Advancement of Cost Engineering International's (ACCE) classification system is a feasibility study with an accuracy of -30 to +50%. These are often referred to as a Rough-Order-of-Magnitude estimates or ROM. Various Cost estimating accuracies are depicted in Table 2 - Cost Estimate Classifications which are based on the level of information available.

(From DOE G 413.3-21A 6-6-2018 Estimating Guide
Table 4-2. Cost Estimate Classification for Process Industries)

ESTIMATE CLASS	Primary Characteristic	Secondary Characteristic		
	MATURITY LEVEL OF PROJECT DEFINITION DELIVERABLES Expressed as % of complete definition	END USAGE Typical purpose of estimate	METHODOLOGY Typical estimating method	EXPECTED ACCURACY RANGE Typical variation in low and high ranges
Class 5	0% to 2%	Concept screening	Capacity factored, parametric models, judgment, or analogy	L: -20% to -50% H: +30% to +100%
Class 4	1% to 15%	Study or feasibility	Equipment factored or parametric models	L: -15% to -30% H: +20% to +50%
Class 3	10% to 40%	Budget authorization or control	Semi-detailed unit costs with assembly level line items	L: -10% to -20% H: +10% to +30%
Class 2	30% to 75%	Control or bid/tender	Detailed unit cost with forced detailed take-off	L: -5% to -15% H: +5% to +20%
Class 1	65% to 100%	Check estimate or bid/tender	Detailed unit cost with detailed take-off	L: -3% to -10% H: +3% to +15%

Table 2 - Cost Estimate Classification¹

¹ AACE International Recommended Practice 18R-97, Cost Estimate Classification System – As Applied in Engineering, Procurement, and Construction for the Process Industries (March 2016).

2. TECHNICAL SCOPE

The scope of these decommissioning estimates is restricted to estimating the decommissioning cost of buildings B17, B70, B56, B54 and B19. It should be noted B18 reflects only soil remediation as the site was an outdoor laydown area for materials and equipment, without significant structures. All areas are located on the site at Lucas Heights (ANSTO).

The cost estimate for each structure includes the cost to complete removal and demolition of buildings to ground level including the removal of radiological or hazardous material. Each building is estimated as a “stand alone” project and does not factor in efficiencies that may be gained by decommissioning buildings in parallel or treating all areas as one project. Except for B18, the estimated decommissioning costs do not include remediation costs for below grade material such as soil, piping, equipment or underground utilities.

The estimates were prepared using a “template” approach where ANSTO can adjust certain parameters for each structure being estimated. The DCE electronic file is an open source excel workbook which contains the following spreadsheets:

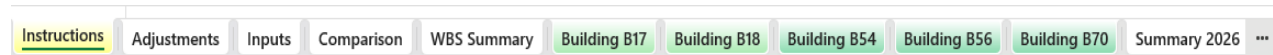


Figure 1- Workbooks Tabs

Each of the workbooks are linked where applicable and provide the following information:

- **Instructions** – This tab provides an overview of the worksheets as well as guidance on how to update an estimate or to prepare a new DCE.
- **Adjustments** – Previously developed estimates for diverse types of facilities, were used to develop average unit rates for activities such as decontamination, clean removal, demolition and site restoration. The base estimates used the same classification as were used by PT&C to provide consistency as shown in Table 2.2 below.

Facility Group	Facility Type	
	Radiological	Non- Radiological
A. Process	A1	A2
B. Laboratory	B1	B2
C. Industrial	C1	C2
D. Non-Industrial	D1	D2

Table 3 - Building Classification Ref 6

CAA used these classifications to develop the ANSTO estimates are from facilities assumed to have a designation of C1, on average. A facility designated as B1 would require an increase in costs, in this case a 50% increase. Facilities with a designation of D1 or D2 would require less effort. Estimators’ judgement was used to arrive at these adjustment factors.

- **Inputs** – This includes generic data that is applicable for of the DCE such as labor rates and waste costs.
- **Comparison** – For information only. This worksheet compares CAA class four decommissioning cost estimates to the 2018 class four estimates prepared by PT&C.
- **WBS Summary** – Provides a Work Breakdown Structure costs by individual area or structure.
- **Buildings “X”** – This worksheet provides the cost breakdown for each individual structure or area. For this report, it includes DCEs for buildings B17, B70, B56, B54 and B19
- **Summary 2026** – This worksheet compiles and merges the U.S. DCE data used to develop unit costs for activities such as decontamination, clean removal, demolition and site restoration. This file ***should not be changed*** since it provides the summary details of the combined US DCEs which are used along with specific ANSTO building parameters to determine costs. Any changes to this worksheet would invalidate the CAA decommissioning cost estimates provided in this report.
- **Hidden Files** – This is the historical data used to develop the unit costs described above. This information should not be changed or modified. As with the summary worksheet, any changes to this worksheet would invalidate the CAA decommissioning cost estimates.

3. METHOD and APPROACH

CAA prepared estimates on five structures or areas selected by ANSTO: B17, B70, B18, B54, & B56. Each structure was estimated as a stand-alone project using the process as described below. These are rough-order-of-magnitude (ROM) estimates based on limited site-specific data available for use. The estimates were developed using site data from the 2018 Project Time and Cost (Ref 6), data analyzed and compiled from U.S. Decommissioning Cost Estimates, and the estimators experience. Using this information, the team developed generic “unit cost” for various activities.

The sequence used in developing the decommissioning cost estimates are as follows:

- a. The cost estimators first reviewed and analyzed various cost estimates they have prepared over the last 15-20 years for both nuclear and non-nuclear facilities. The U.S. estimates were selected based on estimator’s judgement as to applicability to this project. The purpose was to identify specific cost estimating data and details which could be analyzed, combined and used to develop a generic “cost/square foot” which could be used for all the ANSTO structures. This approach factored in specific building DCE attributes such as clean and contaminated systems and structures removal, asbestos removal and final status survey costs.
- b. The US estimates used for this analysis included ten individual estimates for five different sites. The ten estimates included laboratories, research reactors, office buildings, hot cells and a commercial nuclear power plant. These estimates included both clean and contaminated areas. From these estimates, an average removal cost per square foot was determined for clean and contaminated systems and structures. The estimates also included asbestos removal, soil removal, hot cell removal and conducting final surveys. The same process was used to determine waste volume/square foot for clean material, contaminated material and hazardous material.
- c. The estimates were also escalated to 2026 US dollars and then converted to 2026 Australian dollars. The unit rates were converted to cost per square meter. *NOTE: This occurs on the Summary 2026 tab where the unit rates are summarized.*

- d. Building specific data, including building area and building use, was entered into the Input tab, along with staff salaries, waste disposal costs, escalation rates and other site-specific information. Included in the input tab are the building designations B1, C1, D1, and D2. These designations align with the PT&C estimates and were used to modify the estimates based on the adjustment factors identified in the Adjustment worksheet (tab). Crew sizes were adjusted as necessary and included in the Input tab. This adjustment accounts for a deviation from the average crew size of 12 determined from the 2018 PT&C estimates.
- e. The ANSTO information and the CAA information were then aligned on the building specific tabs. Total removal costs and waste volumes were determined on the building specific tabs. The waste volume was also used to calculate the waste disposal cost where appropriate. At this time, there are no costs associated with the disposal or storage of LLW or ILW. The building specific tabs also include management, equipment and consumable, and planning costs.
- f. Costs are then summarized on the WBS Summary tab based on the Work Breakdown Structure (WBS) shown below in Figure 2.

WBS	Description
1.0	Building B17
1.1	Project Management
1.1.1	Staff
1.1.2	Equipment, Materials & Consumables
1.2	Project Planning
1.2.1	Procedure Development
1.2.2	QA and Safety Documents
1.2.3	Mobilization
1.3	Preliminary Site Investigation
1.3.1	Decon and RR
1.3.2	Site Prep
1.4	Detailed Characterization
1.5	Decontamination
1.5.1	Asbestos Removal
1.5.2	Remove systems and Components
1.5.3	Decontaminate Structure
1.5.4	Final Survey
1.6	Site Remediation
1.6.1	Remove Building
1.6.2	Grade and Seed
1.6.3	Backfill
1.7	Waste Disposal

Figure 2 - Work Breakdown Structure Example

4. QUALIFICATIONS, ASSUMPTIONS and EXCLUSIONS

This section describes the key project qualifications and cost assumptions that, according to the DOE's Cost Estimating Guide, provide a "bounding" of the estimate and scope. Specifically, this section describes the assumed condition(s) under which the estimators believe the project work scope will be performed.

It should be noted the estimates were prepared with no site visit and limited information as to the specific type and amount of material being removed from each building such as hazardous and radioactive material, piping and equipment. Without detailed site-specific information, the estimates were prepared

using limited data focusing on the data from the 2018 PT&C report (Ref 6) along with CAA's estimating experience from decommissioning cost estimates prepared over the last 15 - 20 years on various types of projects. The unit cost factors (UCF) and estimating method and approach used by CAA for this project was developed specifically for ANSTO use. The CAA team that prepared these estimates have over 100 years of combined nuclear operations, decommissioning, and cost estimating experience.

4.1 Assumptions:

The unit costs for the development of the decommissioning cost estimates came primarily from information obtained from previous US decommissioning cost estimates. The assumptions and exclusions shown below are generic assumptions that are still applicable based on the U.S. DCEs used by CAA in preparing the estimates. This estimate includes costs for the following:

- Each building is treated as a stand-alone project and does not factor in efficiencies of grouping specific structures based on location, type of facility or whether it contains radioactive or hazardous material.
- The cost estimate is limited to removal of all structures and material above grade and except for B18, which is soil removal only. It is assumed there is no soil removal required for any other structure.
- This estimate does not include the storage/disposal cost of LLW or ILW.
- This estimate does include the cost for the storage/disposal of Industrial and Hazardous waste at the rates provided by ANSTO.
- Management oversight and planning cost are included.
- The determination of the type of structures and contamination level were extracted from the 2018 PT&C decommissioning cost estimate report (Ref 6).
- The D&D cost estimates are provided in 2026, Australian dollars.
- Australian escalation factors are used to escalate the base US estimates to 2026 dollars.
- As with U.S. DCEs, it is assumed all contractors are qualified and licensed. Personnel are professionally trained and qualified and meet all on site requirements such as conducting asbestos removal or radiological surveys.
- The Contractor does not experience any delays, work slowdown, or stoppage that is outside of its control.
- Costs for Contractor licenses and permits are not included in the remediation estimate.
- An estimated duration for completing each building being decommissioned was determined based on estimator's experience along with ANSTO input on previously completed projects. This estimate also considered the size of the building by square meters or "footprint" of the building along with type of material expected to be remediated.

4.2 Exclusions

- The only Historical Site Assessment (HSA) was prepared for B17. No other HSAs or detailed site characterization reports have been performed at the time of this estimate.

- No utility costs such as water or electricity are included but are typically included into the U.S. DCEs used for this project.
- Costs are included in isolating each building from the site electrical and plumbing systems. No utility operating or relocation/modification costs for electricity, water, sewage, irrigation, telephone, or cable lines are included.
- Demolition or removal of surface structures such as roads, concrete driveways and sidewalks are not factored into the DCE.
- Collateral costs such as outside support services, fees, taxes have not been allocated or assigned to the estimate.
- A risk analysis has not been performed.
- The cost estimate does not include severance packages or other compensation for ANSTO employees.
- Regulatory oversight costs or costs associated with regulatory interface or public hearings are not included.
- Except for B18, no soil remediation or removal of underground material, structures, or utilities are included.
- Detailed building inventory or quantities of hazardous or radioactive material are not included.

5. BUILDING DESCRIPTIONS

Cost estimates for decommissioning were prepared individually for each of the five buildings: B17, B70, B18, B54, and B56 as described below. The buildings are all located on the ANSTO Lucas Height Site (Figure 3). Information provided below was extracted from the Project Time and Cost September 14, 2018 Report “Cost Estimate for the Lucas Heights Site Decommissioning and Dismantlement (D&D)” (Ref. 6). For the purposes of this estimate, building 17A and 17B are being combined and designated as Building 17.



Figure 3 - ANSTO Lucas Heights

- **B17A** - Nuclear Services (Material Storage - designated as Building B17A) has a gross area of 704 m². The facility is considered contaminated and is constructed of a steel frame with Steel Truss Roof and Metal Deck Diaphragm Wall Construction. Note B17A & B are combined in this report.
- **B17B** - Nuclear Operations (Workshop) (designated as Building B17B) has a gross area of 158 m². The facility usage was for waste processing or handling of radioactive material and has a construction type designation of Steel Frame Framing with Steel Truss Roof and Metal Deck Diaphragm Wall Construction. Note B17A & B are combined in this report.
- **B18** – The was a soil remediation project and the area was used as an outdoor laydown area. This estimate is for remediation of the soil and asphalt. The total area of remediation is assumed to be 6,900 m² to a depth of 1 meter.
- **B54** - ANSTO Health (Mo99 Processing Facility) and has a gross area of 1,204 m². The facility usage is as a nuclear chemical processing facility and is considered chemically and radiologically contaminated. It is constructed of precast Concrete Framing with Precast Concrete Deck Roof and Precast/Tilt-up Concrete Wall.
- **B56** – This is the Institute of Materials & Engineering Science and has a gross area of 1,557 m². The facility houses Specialized Laboratories, including Clean Rooms, Technical Workshops and office areas. It is constructed of Concrete/Brick Frame and Brick/Glass Façade The end state for the facility is complete decontamination and decommissioning. This facility is considered radiologically contaminated.
- **B70** – This is the High Flux Australian Reactor (HIFAR) electrical power supply building consisting of duplicated rooms for A and B systems housing the two-diesel generator sets and electrical switchboards, UPS equipment, batteries, rectifiers and inverters.

6. BASIS OF ESTIMATE

These estimates are considered class IV estimates used for feasibility studies. In accordance with the Associations for the Advancement of Cost Engineering International (AACEI), Class IV estimates are a concept or feasibility study (often referred to as a rough of Order of Magnitude estimate or ROM). Consistent with such expectations, this estimate was developed based on limited data with no site walkdowns, detailed building inventory and limited or no building characterization or historical site assessment data.

The elements in the basis of estimates are included as part of the overall adjustments in the baseline estimates used to identify unit costs. For this DCE approach, CAA did not breakout or analyze the various Basis of Estimate elements since they are already factored into the U.S. DCE.

6.1 Risk Basis

A qualitative or quantitative risk analysis has not been performed and is not included or factored into these estimates. The estimates were prepared with limited detailed information such as site/building characterization, Historical Site Assessment or information specifically on how much material will need to be dismantled and disposed of (clean, hazardous and radioactive).

The risks associated with executing the project based on these ROM estimates are primarily due to a lack of site characterization. This lack of detailed information could result in increased costs due to the amount of material, equipment and radioactive or hazardous material assumed in each facility. It is highly recommended that in keeping with industry practice, a more detailed cost estimate and risk analysis should be carried out prior to physical decommissioning.

6.2 Cost Basis

These cost estimates were prepared using limited information provided by ANSTO, analyzing previous United States (US) cost estimates, and the estimating team's knowledge and experience. The U.S. estimates used as a baseline for these estimates included laboratories, research reactors, office buildings, hot cells and a commercial nuclear power plant. These estimates also include both clean and contaminated areas. From these estimates, an average cost per square foot was determined for clean and contaminated systems removal, clean and contaminated structure removal, asbestos removal, soil removal, hot cell removal, and final site survey. The same process was used to determine waste volume/square foot for clean material, contaminated material and hazardous material.

7. SPREADSHEETS

The estimates were prepared using a "template" approach which allows ANSTO to adjust certain parameters for each structure being estimated. The DCE electronic file is a set of Microsoft Excel source workbooks. Refer to section 2 Technical Scope for a description of each spreadsheet.

8. DECOMMISSIONING COST ESTIMATES – BY BUILDINGS

Below is a brief description and a DCE for each building. These buildings are all located on the ANSTO Lucas Height Site.

8.1 Building B17 (B17A and 17B)

Nuclear Services (Material Storage) and Nuclear Operations (Workshop). The combined gross area of 862 m².

WBS	Description	LSA Waste (M3)	Hazardous Waste Vol (M3)	Mixed Waste Vol (M3)	Clean Waste Vol. (M3)	Mat's, Consumables & Equip.	Labor Cost	LSA Waste Cost (TBD)	Hazardous Waste Cost	Mixed Waste Cost	Clean Waste Cost	Total Cost
1.0	Building B17											
1.1	Project Management											
1.1.1	Staff						268,756					\$ 268,756
1.1.2	Equipment, Materials & Consumables					677,458						\$ 677,458
1.2	Project Planning											
1.2.1	Procedure Development						22,935					\$ 22,935
1.2.2	QA and Safety Documents						22,935					\$ 22,935
1.2.3	Mobilization						123,379					\$ 123,379
1.3	Preliminary Site Investigation											\$ -
1.3.1	Decontamination Readiness Review						See Note					\$ -
1.3.2	Site Prep						45,403					\$ 45,403
1.4	Detailed Characterization						74,430					\$ 74,430
1.5	Decontamination											\$ -
1.5.1	Asbestos Removal						0					\$ -
1.5.2	Remove systems and Components						447,634					\$ 447,634
1.5.3	Decontaminate Structure	20	0	2	190		192,193					\$ 192,193
1.5.4	Final Survey						208,900					\$ 208,900
1.6	Site Remediation				664							\$ 392,415
1.6.1	Remove Building						392,415					\$ 392,415
1.6.2	Grade and Seed						27,281					\$ 27,281
1.6.3	Backfill						62,496					\$ 62,496
1.7	Waste Disposal								2,458		452,853	\$ 455,311
	TOTAL	20	0	2	855	677,458	1,888,759	0	2,458	0	452,853	3,021,528

Figure 4 - B17 Cost Estimate

8.2 Area B18

There was a soil remediation project and the area was used as an outdoor laydown area. This estimate is for remediation of the soil and asphalt.

WBS	Description	LSA Waste (M3)	Hazardous Waste Vol (M3)	Mixed Waste Vol (M3)	Clean Waste Vol. (M3)	Mat's, Consumables & Equip.	Labor Cost	LSA Waste Cost (TBD)	Hazardous Waste Cost	Mixed Waste Cost	Clean Waste Cost	Total Cost
1.0	Building B18											
1.1	Project Management											
1.1.1	Staff						134,378					\$ 134,378
1.1.2	Equipment, Material & Consumables					465,732						\$ 465,732
1.2	Project Planning											
1.2.1	Procedure Development						22,935					\$ 22,935
1.2.2	QA and Safety Documents						22,935					\$ 22,935
1.2.3	Mobilization						123,379					\$ 123,379
1.3	Preliminary Site Investigation											\$ -
1.3.1	Decontamination Readiness Review						See Note 1					\$ -
1.3.2	Site Prep						44,658					\$ 44,658
1.4	Detailed Characterization						297,722					\$ 297,722
1.5	Decontamination											\$ -
1.5.1	Remove Soil and Asphalt	3,450			3,450		294,144					\$ 294,144
1.5.2	Final Survey						1,672,173					\$ 1,672,173
1.6	Site Remediation											\$ -
1.6.1	Remove Building						N/A					\$ -
1.6.2	Grade and Seed						109,189					\$ 109,189
1.6.3	Backfill						164,128					\$ 164,128
1.7	Waste Disposal										1,827,537	\$ 1,827,537
	TOTAL	3,450	0	0	3,450	465,732	2,885,641	0	0	0	1,827,537	5,178,910

Figure 5 - B18 Cost Estimate

8.3 Building B54

ANSTO Health (Mo99 Processing Facility) and has a gross area of 1,204 SM. The facility usage is as a nuclear chemical processing facility and is considered chemically and radiologically contaminated. It is constructed of precast Concrete Framing with Precast Concrete Deck Roof and Precast/Tilt-up Concrete Wall.

WBS	Description	LSA Waste (M3)	Hazardous Waste Vol (M3)	Mixed Waste Vol (M3)	Clean Waste Vol. (M3)	Mat's, Consumables & Equip.	Labor Cost	LSA Waste Cost (TBD)	Hazardous Waste Cost	Mixed Waste Cost	Clean Waste Cost	Total Cost
1.0	Building B54											
1.1	Project Management											
1.1.1	Staff						614,958					\$ 614,958
1.1.2	Equipment, Material & Consumables					1,975,720						\$ 1,975,720
1.2	Project Planning											
1.2.1	Procedure Development						22,935					\$ 22,935
1.2.2	QA and Safety Documents						22,935					\$ 22,935
1.2.3	Mobilization						225,124					\$ 225,124
1.3	Preliminary Site Investigation											\$ -
1.3.1	Decontamination Readiness Review						See Note 1					\$ -
1.3.2	Site Prep						44,658					\$ 44,658
1.4	Detailed Characterization						446,582					\$ 446,582
1.5	Decontamination	30	329	3	266							
1.5.1	Asbestos Removal						880,073					\$ 880,073
1.5.2	Remove systems and Components						3,751,404					\$ 3,751,404
1.5.3	Decontaminate Structure						2,123,536					\$ 2,123,536
1.5.4	Final Survey						583,564					\$ 583,564
1.6	Site Remediation				1,075							
1.6.1	Remove Building						1,644,320					\$ 1,644,320
1.6.2	Remove Hot Cell						544,169					\$ 544,169
1.6.3	Grade and Seed						76,211					\$ 76,211
1.6.4	Backfill						174,584					\$ 174,584
1.7	Waste Disposal								290,166	27,903	710,467	\$ 1,028,535
	TOTAL	30	329	3	1,341	1,975,720	11,155,053	0	290,166	27,903	710,467	14,159,309

Figure 6 - B54 Cost Estimate

8.4 Building B56

This is the Advanced Material Program Advanced and has a gross area of 1,557 SM. The facility houses Specialized Laboratories, including Clean Rooms, Technical Workshops and office areas. It is constructed of Concrete/Brick Frame and Brick/Glass Façade The end state for the facility is complete decontamination and decommissioning. This facility is considered radiologically contaminated.

WBS	Description	LSA Waste (M3)	Hazardous Waste Vol (M3)	Mixed Waste Vol (M3)	Clean Waste Vol. (M3)	Mat's, Consumables & Equip.	Labor Cost	LSA Waste Cost (TBD)	Hazardous Waste Cost	Mixed Waste Cost	Clean Waste Cost	Total Cost
1.0	Building B56											
1.1	Project Management											
1.1.1	Staff						256,233					\$ 256,233
1.1.2	Equipment, Material & Consumables					1,843,800						\$ 1,843,800
1.2	Project Planning											
1.2.1	Procedure Development						22,935					\$ 22,935
1.2.2	QA and Safety Documents						22,935					\$ 22,935
1.2.3	Mobilization						225,124					\$ 225,124
1.3	Preliminary Site Investigation											\$ -
1.3.1	Decontamination Readiness Review						See Note 1					\$ -
1.3.2	Site Prep						44,658					\$ 44,658
1.4	Detailed Characterization						297,722					\$ 297,722
1.5	Decontamination	39	425	4	344							
1.5.1	Asbestos Removal						0					\$ -
1.5.2	Remove systems and Components						4,851,275					\$ 4,851,275
1.5.3	Decontaminate Structure						2,082,904					\$ 2,082,904
1.5.4	Final Survey						754,659					\$ 754,659
1.6	Site Remediation				1,391							
1.6.1	Remove Building						2,126,417					\$ 2,126,417
1.6.2	Grade and Seed						98,555					\$ 98,555
1.6.3	Backfill						225,770					\$ 225,770
1.7	Waste Disposal								375,240	3,961	918,768	\$ 1,297,969
	TOTAL	39	425	4	1,734	1,843,800	11,009,187	0	375,240	3,961	918,768	14,150,956

Figure 7 - B56 Cost Estimate

8.5 Building B70

This is a HIFAR electrical power supply building consisting of duplicated rooms for A and B systems housing the two-diesel generator sets and electrical switchboards, UPS equipment, batteries, rectifiers and inverters. Connected by trenches to Sub-Station No.1 and Building 15 and the ECR.

WBS	Description	LSA Waste (M3)	Hazardous Waste Vol (M3)	Mixed Waste Vol (M3)	Clean Waste Vol (M3)	Mat's, Consumables & Equip.	Labor Cost	LSA Waste Cost (TBD)	Hazardous Waste Cost	Mixed Waste Cost	Clean Waste Cost	Total Cost
1.0	Building B70											
1.1	Project Management											
1.1.1	Staff						71,962					\$ 71,962
1.1.2	Equipment, Material & Consumables					188,018						\$ 188,018
1.2	Project Planning											
1.2.1	Procedure Development						11,468					\$ 11,468
1.2.2	QA and Safety Documents						11,468					\$ 11,468
1.2.3	Mobilization						32,130					\$ 32,130
1.3	Preliminary Site Investigation											\$ -
1.3.1	Decontamination Readiness Review						See Note 1					\$ -
1.3.2	Site Prep						44,658					\$ 44,658
1.4	Detailed Characterization						0					\$ -
1.5	Decontamination	0	0	0	0							
1.5.1	Asbestos Removal						165,014					\$ 165,014
1.5.2	Remove systems and Components						35,618					\$ 35,618
1.5.3	Decontaminate Structure		42		34		0					\$ -
1.5.4	Final Survey						27,666					\$ 27,666
1.6	Site Remediation				0							
1.6.1	Remove Building						25,985					\$ 25,985
1.6.2	Remove Hot Cell						0					\$ -
6.2	Grade and Seed						3,613					\$ 3,613
1.6.3	Backfill						8,277					\$ 8,277
1.7	Waste Disposal								36,684	12,807	89,820	\$ 139,311
TOTAL		0	42	0	34	188,018	437,859	0	36,684	12,807	89,820	765,187

Figure 8 - B70 Cost Estimate

9. WBS DICTIONARY

For purposes of cost estimating, the work is organized and decomposed according to the WBS dictionary for each building as shown below.

WBS Dictionary		
Level	WBS Code	Definition
1	Building B17	
1.1	Project Management	
1.1.1	Staff	Includes the management staff as well as decom technician/Safety SMI
1.1.2	Equipment, Materials & Consumables	Includes various large and small equipment, PPEs as well as Rad instrumentation.
1.2	Project Planning	
1.2.1	Procedure Development	Develops the procedure necessary to decommission the facility
1.2.2	QA and Safety Documents	QA oversight of the project activities and development of safety documents
1.2.3	Mobilization	Includes staging equipment, setting up work areas/office and training
1.3	Preliminary Site Investigation	
1.3.1	Decontamination Readiness Review	The readiness review contractor staff is included in undistributed costs.
1.3.2	Site Prep	This includes initial site surveys, setting up of work areas, installing barriers and HEPA and setting up staging areas. Also includes costs for setting up radiation segregation station, isolating the building including electrical and piping components.
1.4	Detailed Characterization	Conduct Site Characterization and issue report
1.5	Decontamination	
1.5.1	Asbestos Removal	No asbestos removal is included in this estimate
1.5.2	Remove systems and Components	Includes the removal of system and components including clean, LLW, Hazardous and Mixed waste.
1.5.3	Decontaminate Structure	Decontamination of site structures/building for free release
1.5.4	Final Survey	Complete final survey to allow removal of structures.
1.6	Site Remediation	
1.6.1	Remove Building	Complete removal of B17 building/structure,
1.6.2	Grade and Seed	Grade and seed an area 2X building footprint
1.6.3	Backfill	Backfill an area assumed to be the same area as the structure and 10 feet deep
1.7	Waste Disposal	Disposal of clean, LLW, Hazardous and mixed waste.

Figure 9 - B17 WBS Dictionary

WBS Dictionary		
WBS	WBS Code	Definition
1.0	Building B18	
1.1	Project Management	
1.1.1	Staff	Includes the management staff as well as decom technician/Safety SMI
1.1.2	Equipment, Material & Consumables	Includes various large and small equipment, PPEs as well as Rad instrumentation.
1.2	Project Planning	
1.2.1	Procedure Development	Develops the procedure necessary to decommission the facility
1.2.2	QA and Safety Documents	QA oversight of the project activities and development of safety documents
1.2.3	Mobilization	Includes staging equipment, setting up work areas/office and training
1.3	Preliminary Site Investigation	
1.3.1	Decontamination Readiness Review	The readiness review contractor staff is included in undistributed costs.
1.3.2	Site Prep	This includes initial site surveys, setting up of work areas, installing barriers and HEPA and setting up staging areas. Also includes costs for setting up radiation segregation station, isolating the building including electrical and piping components.
1.4	Detailed Characterization	Conduct Site Characterization and issue report
1.5	Decontamination	
1.5.1	Remove Soil and Asphalt	Includes the removal contaminated soil and asphalt
1.5.2	Final Survey	Complete final survey to allow removal of structures.
1.6	Site Remediation	
1.6.1	Remove Building	Complete removal of B17 building/structure,
1.6.2	Grade and Seed	Grade and seed an area 2X building footprint
1.6.3	Backfill	Backfill an area assumed to be the same area as the structure and 10 feet deep
1.7	Waste Disposal	Disposal of clean and LLW

Figure 10 - B18 WBS Dictionary

WBS Dictionary		
WBS	WBS Code	Definition
1.0	Building B54	
1.1	Project Management	
1.1.1	Staff	Includes the management staff as well as decom technician/Safety SMI
1.1.2	Equipment, Material & Consumables	Includes various large and small equipment, PPEs as well as Rad instrumentation.
1.2	Project Planning	
1.2.1	Procedure Development	Develops the procedure necessary to decommission the facility
1.2.2	QA and Safety Documents	QA oversight of the project activities and development of safety documents
1.2.3	Mobilization	Includes staging equipment, setting up work areas/office and training
1.3	Preliminary Site Investigation	
1.3.1	Decontamination Readiness Review	The readiness review contractor staff is included in undistributed costs.
1.3.2	Site Prep	This includes initial site surveys, setting up of work areas, installing barriers and HEPA and setting up staging areas. Also includes costs for setting up radiation segregation station, isolating the building including electrical and piping components.
1.4	Detailed Characterization	Conduct Site Characterization and issue report
1.5	Decontamination	
1.5.1	Asbestos Removal	Remove asbestos tile and contaminated asbestos on piping and components
1.5.2	Remove systems and Components	Includes the removal of system and components including clean, LLW, Hazardous and Mixed waste.
1.5.3	Decontaminate Structure	Decontamination of site structures/building for free release
1.5.4	Final Survey	Complete final survey to allow removal of structures.
1.6	Site Remediation	
1.6.1	Remove Building	Remove structure to grade
1.6.2	Remove Hot Cell	Remove Hot Cell after Decontaminated
1.6.3	Grade and Seed	Grade and seed an area 2X building footprint
1.6.4	Backfill	Backfill an area assumed to be the same area as the structure and 10 feet deep
1.7	Waste Disposal	Disposal of clean, LLW, Hazardous and mixed waste.

Figure 11 - B54 WBS Dictionary

WBS Dictionary		
WBS	WBS Code	Definition
1.0	Building B56	
1.1	Project Management	
1.1.1	Staff	Includes costs for 3 managers and a decom technician/Safety SMI
1.1.2	Equipment, Material & Consumables	Includes various large and small equipment, PPEs as well as Rad instrumentation.
1.2	Project Planning	
1.2.1	Procedure Development	Two personnel to develop procedure necessary to complete the project.
1.2.2	QA and Safety Documents	Two personnel to provide QA oversight and development of safety documents
1.2.3	Mobilization	Historical estimate averages are based on a work crew of 12.
1.3	Preliminary Site Investigation	
1.3.1	Decontamination Readiness Review	The readiness review contractor staff is included in undistributed costs.
1.3.2	Site Prep	This includes initial site surveys, setting up of work areas, installing barriers and HEPA and setting up staging areas. Also includes costs for setting up radiation segregation station, isolating the building including electrical and piping components.
1.4	Detailed Characterization	Conduct Site Characterization and issue report
1.5	Decontamination	
1.5.1	Asbestos Removal	Remove asbestos tile and contaminated asbestos on piping and components
1.5.2	Remove systems and Components	Remove clean, LLW, Hazardous and mixed waste
1.5.3	Decontaminate Structure	Decontamination includes Hot Cells, building, and removal of hazardous material
1.5.4	Final Survey	Cost to complete final survey to allow removal of structures.
1.6	Site Remediation	
1.6.1	Remove Building	Remove structure to grade
1.6.2	Remove Hot Cell	Remove Hot Cell after Decontaminated
1.6.3	Backfill	Backfill (assume building footprint x 10' deep)
1.7	Waste Disposal	All waste include clean, hazardous, mixed and LLW

Figure 12 - B56 WBS Dictionary

WBS Dictionary		
WBS	WBS Code	Definition
1.0	Building B70	
1.1	Project Management	
1.1.1	Staff	Includes the management staff as well as decom technician/Safety SMI
1.1.2	Equipment, Material & Consumables	Includes various large and small equipment, PPEs as well as Rad instrumentation.
1.2	Project Planning	
1.2.1	Procedure Development	Develops the procedure necessary to decommission the facility
1.2.2	QA and Safety Documents	QA oversight of the project activities and development of safety documents
1.2.3	Mobilization	Includes staging equipment, setting up work areas/office and training
1.3	Preliminary Site Investigation	
1.3.1	Decontamination Readiness Review	The readiness review contractor staff is included in undistributed costs.
1.3.2	Site Prep	This includes initial site surveys, setting up of work areas, installing barriers and HEPA and setting up staging areas. Also includes costs for setting up radiation segregation station, isolating the building including electrical and piping components.
1.4	Detailed Characterization	Conduct Site Characterization and issue report
1.5	Decontamination	
1.5.1	Asbestos Removal	Remove asbestos tile and contaminated asbestos on piping and components
1.5.2	Remove systems and Components	Remove clean, LLW, Hazardous and mixed waste
1.5.3	Decontaminate Structure	Remove clean, LLW, Hazardous and mixed waste
1.5.4	Final Survey	Cost to complete final survey to allow removal of structures.
1.6	Site Remediation	
1.6.1	Remove Building	Remove structure to grade
1.6.2	Remove Hot Cell	Remove Hot Cell after Decontaminated
1.6.3	Grade and Seed	Backfill (assume building footprint x 10' deep)
1.6.4	Backfill	All waste include clean, hazardous, mixed and LLW
1.7	Waste Disposal	All waste include clean, hazardous, mixed and LLW

Figure 13 - B70 WBS Dictionary

10. REFERENCES

The material listed below was reviewed and information extracted from the Australian Nuclear Science and Technology Organisation - Cost Estimate for the Lucas Heights Site Decommissioning and Dismantlement (D&D) - 14 September 2018 - Project Time & Cost, LLC

Other material used as guidance for developing cost estimates included the following:

1. Minor Consultancy Terms and Conditions AF-1684 (ABN 47 956 969 590)
2. AIF - Atomic Industrial Forum, Inc., AIF/NESP-036, "Guidelines for Producing Nuclear Power Plant Decommissioning Cost Estimates," May 1986.
3. AACE International Recommended Practice No. 17-R-97, "Cost Estimate Classification System," 2003
4. AACE - American Association of Cost Engineers, "Project Cost Engineers Handbook," 2nd Edition
5. Project Management Book of Knowledge (PMBOK) – 6th Edition